

**A Project Report On
Greatkart Ecommerce**

**SUBMITTED FOR THE PARTIAL FULFILLMENT OF THE
MASTER'S DEGREE IN COMPUTER APPLICATION
FOR THE YEAR -2025-26**

SUBMITTED TO



**KN University
S.G. Highway, Gota, Ahmedabad, Gujarat**

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THE GUIDANCE OF UNDER

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&
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ACKNOWLEDGEMENT

We are very happy to present our project work titled Greatkart Ecommerce, developed using Python and Django. This project was a great learning experience for us, and we are thankful to everyone who supported and guided us during its completion.

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Lastly, we are deeply thankful to our parents and family members for their continuous support, encouragement, and motivation. Their blessings and confidence always inspired us to complete this project successfully.

Name of Student
Hitesh Sathvara

PREFACE

The project Greatkart Ecommerce has been developed as an academic project using Python and Django. The purpose of this project is to understand and implement the concepts of web application development in a practical manner. It is designed mainly for educational and learning purposes.

This project includes important features of an e-commerce system such as user registration, login, product management, shopping cart, order management, and admin control. Through the development of this project, we learned various concepts like frontend and backend integration, database connectivity, authentication, session handling, and CRUD operations using the Django framework.

The main aim of this project is to gain practical knowledge of modern web technologies and improve programming, problem-solving, and project development skills. It also helped us understand how real-world web applications are designed and managed.

Working on this project gave us valuable hands-on experience and increased our confidence in developing dynamic web applications. We hope this project will serve as a useful academic reference for students and beginners who want to learn web development using Python and Django.

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1.INTRODUCTION

Nowadays, web applications are used everywhere in daily life. E-commerce websites help users to view products, manage orders, and perform shopping activities easily. To learn how such systems work practically, we developed an academic project named Greatkart Ecommerce using Python and Django.

This project is made for learning and educational purposes. It helps us understand how a real web application is developed using frontend and backend technologies. The project includes basic e-commerce features like user registration, login, product display, shopping cart, order management, and admin panel.

While developing this project, we learned many important concepts such as database connectivity, authentication system, session management, CRUD operations, and web page designing. This project also improved our programming knowledge, teamwork, and problem-solving skills.

The main purpose of this project is to gain practical knowledge of web development using Django Framework and understand the working process of an e-commerce website.

2. About Project

Greatkart E-commerce is a web-based academic project developed using Python and the Django Framework. The project is designed to understand the basic working of an e-commerce system in a simple and practical way.

In this project, users can create an account, log in, view products, add products to the cart, and place orders. The admin can manage products, categories, users, and orders through the admin panel.

The project has a simple and user-friendly design so that anyone can use it easily. It is developed mainly for educational purposes and practical learning of web development concepts.

2.1 Existing System

1. Manual Management Process

In many academic projects and small systems, product details, customer records, and orders are managed manually or using basic applications. Data is often stored in registers, Excel files, or separate systems, which makes management difficult.

2. Limited Learning and Practical Features

Traditional academic systems provide only basic functionality and do not offer proper practical understanding of real-world e-commerce concepts like cart management, user authentication, and order handling.

3. Time-Consuming Work

Managing product information, updating records, and handling user data manually requires more time and increases the chances of errors.

4. Lack of Proper User Management

Many existing systems do not provide secure login systems, proper database management, and organized interfaces for users and administrators.

2.2 Proposed System

1. Overview

The Greatkart E-commerce project is a web-based academic project developed using Python and the Django Framework. The main purpose of this project is to learn and understand the practical implementation of web application development and e-commerce system concepts.

2. System Architecture

- **Frontend:** HTML, CSS, Bootstrap, JavaScript
- **Backend:** Python (Django Framework)
- **Database:** SQLite / MySQL

3. Key Features

User Module

- User Registration and Login
- Secure Authentication System
- View Products and Categories
- Search Product Functionality
- Add Products to Shopping Cart
- Simple Order Placement System
- User-Friendly Interface

Administrator Module

- Admin Login System
- Add, Update, and Delete Products
- Manage Product Categories
- Manage User Records
- Manage Orders and Product Details

3.Requirement Analysis: -

3.1 Hardware Requirements:

-  **Processor rand:** Intel
-  **System Type:** 64-bit Operating System
-  **Processor Speed:** 2.42 GHz or above
-  **RAM Size:** 8.00 GB
-  **Hard Disk Size:** 256 GB

3.2 Software Requirements:

-  **Operating System:** Windows 11
-  **Application Server:** Django Development Server
-  **Frontend Technologies:** HTML, CSS, JavaScript, Bootstrap
-  **Backend Technology:** Python with Django Framework
-  **Database:** SQLite

3.3 Data Dictionary: -

store_reviewrating				
Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
subject	VARCHAR	100	NOT NULL	Review headline
review	TEXT	var	NOT NULL	Full review text body
rating	REAL	8	NOT NULL	Star rating (1.0–5.0)
ip	VARCHAR	20	NOT NULL	Reviewer IP address
status	BOOL	1	NOT NULL	Moderation status flag
created_at	DATETIME	8	NOT NULL	Review creation timestamp
updated_at	DATETIME	8	NOT NULL	Last update timestamp
product_id	BIGINT	4	FOREIGN KEY → store_product.id	
user_id	BIGINT	4	FOREIGN KEY → accounts_account.id	

store_product				
Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
product_name	VARCHAR	200	NOT NULL	Full product display name
slug	VARCHAR	200	NOT NULL	URL-safe product identifier
description	TEXT	var	NOT NULL	Product description text
price	INTEGER	4	NOT NULL	Base price in store currency
images	VARCHAR	100	NOT NULL	Path to primary product image
stock	INTEGER	4	NOT NULL	Available inventory count
is_available	BOOL	1	NOT NULL	Listed for purchase flag
created_date	DATETIME	8	NOT NULL	Product creation timestamp
modified_date	DATETIME	8	NOT NULL	Last modification timestamp
category_id	BIGINT	4	FOREIGN KEY → category_category.id	



orders_payment

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
payment_id	VARCHAR	100	NOT NULL	Gateway transaction ID
payment_method	VARCHAR	100	NOT NULL	Payment provider or method
amount_paid	VARCHAR	100	NOT NULL	Amount charged (from gateway)
status	VARCHAR	100	NOT NULL	Payment status string
created_at	DATETIME	8	NOT NULL	Payment creation timestamp
user_id	BIGINT	4	FOREIGN KEY → accounts_account.id	



orders_orderproduct

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
quantity	INTEGER	4	NOT NULL	Number of units ordered
product_price	REAL	8	NOT NULL	Price snapshot at purchase time
ordered	BOOL	1	NOT NULL	Part of a confirmed order
created_at	DATETIME	8	NOT NULL	Record creation timestamp
updated_at	DATETIME	8	NOT NULL	Record last-updated timestamp
order_id	BIGINT	4	FOREIGN KEY → orders_order.id	
payment_id	INTEGER	4	FOREIGN KEY → orders_payment.id	
product_id	INTEGER	4	FOREIGN KEY → store_product.id	
user_id	BIGINT	4	FOREIGN KEY → accounts_account.id	



orders_order

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
order_number	VARCHAR	20	NOT NULL	Human-readable order reference
first_name	VARCHAR	50	NOT NULL	Customer first name
last_name	VARCHAR	50	NOT NULL	Customer last name
phone	VARCHAR	15	NOT NULL	Customer phone number
email	VARCHAR	50	NOT NULL	Customer email address
address_line_1	VARCHAR	50	NOT NULL	Delivery address line 1
address_line_2	VARCHAR	50	NOT NULL	Delivery address line 2
country	VARCHAR	50	NOT NULL	Delivery country
state	VARCHAR	50	NOT NULL	Delivery state
city	VARCHAR	50	NOT NULL	Delivery city
order_note	VARCHAR	100	NOT NULL	Optional delivery instructions
order_total	REAL	8	NOT NULL	Total cost including tax
tax	REAL	8	NOT NULL	Applied tax amount
status	VARCHAR	10	NOT NULL	Order status (New/Accepted/etc.)
ip	VARCHAR	20	NOT NULL	Customer IP at order time
is_ordered	BOOL	1	NOT NULL	True once order is confirmed
created_at	DATETIME	8	NOT NULL	Order creation timestamp
updated_at	DATETIME	8	NOT NULL	Last update timestamp
payment_id	INTEGER	4	FOREIGN KEY → orders_paymentid	
user_id	BIGINT	4	FOREIGN KEY → accounts_accountid	



category_category

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
category_name	VARCHAR	50	NOT NULL	Display name of the category
slug	VARCHAR	100	NOT NULL	URL-safe unique identifier
description	TEXT	var	NOT NULL	Category description
cat_image	VARCHAR	100	NOT NULL	Path to category thumbnail



carts_cartitem_variations

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
cartitem_id	INTEGER	4	FOREIGN KEY → carts_cartitem.id	
variation_id	INTEGER	4	FOREIGN KEY → store_variation.id	



carts_cartitem

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
quantity	INTEGER	4	NOT NULL	Number of units selected
is_active	BOOL	1	NOT NULL	Whether the cart item is active
cart_id	BIGINT	4	FOREIGN KEY → carts_cartId	
product_id	INTEGER	4	FOREIGN KEY → store_product.id	
user_id	BIGINT	4	FOREIGN KEY → accounts_accountid	



carts_cart

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
cart_id	VARCHAR	250	NOT NULL	Unique session cart identifier
date_added	DATE	8	NOT NULL	Date cart was created



accounts_userprofile

Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
address_line_1	VARCHAR	100	NOT NULL	Primary address line
address_line_2	VARCHAR	100	NOT NULL	Secondary address line
city	VARCHAR	20	NOT NULL	City name
state	VARCHAR	20	NOT NULL	State or province
country	VARCHAR	20	NOT NULL	Country name
user_id	BIGINT	4	FOREIGN KEY → accounts_accountid	
profile_picture	VARCHAR	100	NOT NULL	Path to profile image

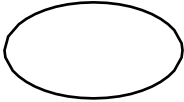
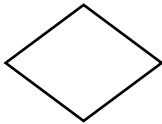
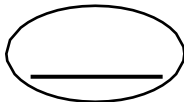
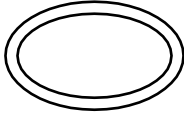


accounts_account

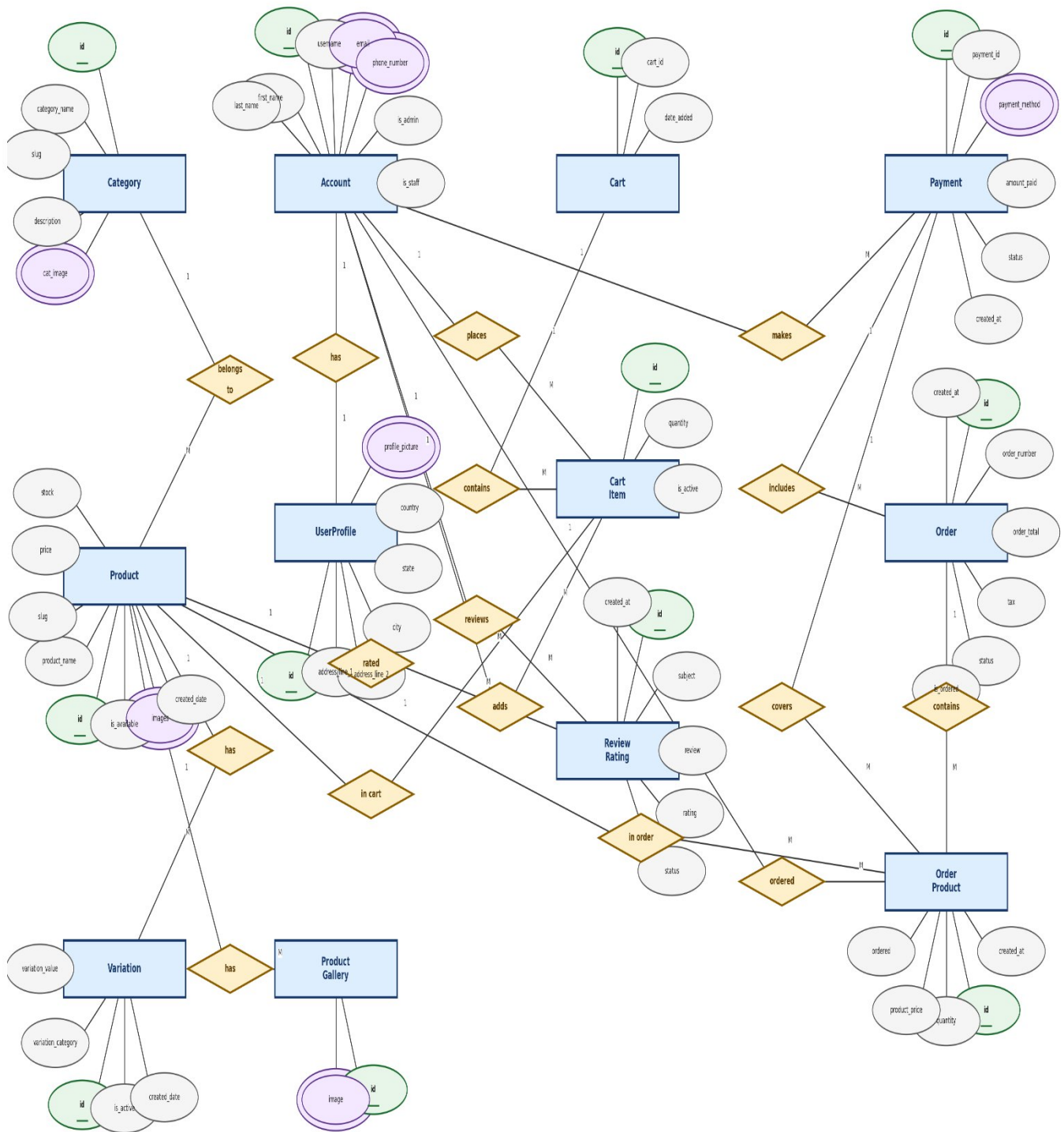
Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key — auto-incremented
password	VARCHAR	128	NOT NULL	Hashed password string
first_name	VARCHAR	50	NOT NULL	User's first name
last_name	VARCHAR	50	NOT NULL	User's last name
username	VARCHAR	50	NOT NULL	Unique login handle
email	VARCHAR	100	NOT NULL	Email used for authentication
phone_number	VARCHAR	50	NOT NULL	Contact phone number
date_joined	DATETIME	8	NOT NULL	Account creation timestamp
last_login	DATETIME	8	NOT NULL	Most recent login timestamp
is_admin	BOOL	1	NOT NULL	Admin access flag
is_staff	BOOL	1	NOT NULL	Can access Django admin
is_active	BOOL	1	NOT NULL	Account active (soft delete)
is_superuser	BOOL	1	NOT NULL	Full superuser permissions

store_variation				
Field Name	Data Type	Size	Constraint	Description
id	INTEGER	4	PRIMARY KEY, AUTO_INCREMENT	Primary key
variation_category	VARCHAR	100	NOT NULL	Attribute type (color, size)
variation_value	VARCHAR	100	NOT NULL	Specific value (Red, XL)
is_active	BOOL	1	NOT NULL	Variation active flag
created_date	DATETIME	8	NOT NULL	Variation creation timestamp
product_id	BIGINT	4	FOREIGN KEY → store_product.id	

ER DIAGRAM SYMBOLS

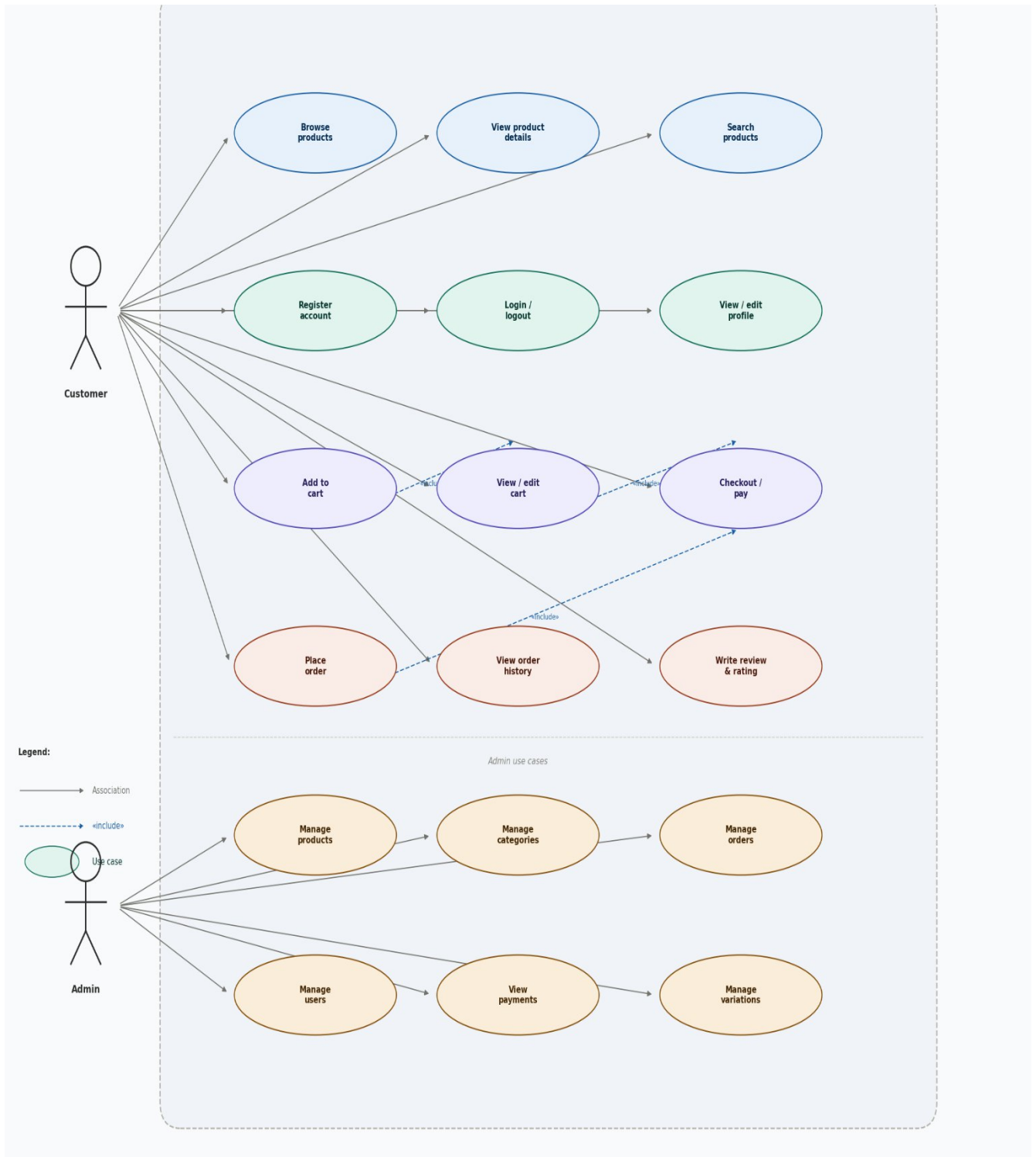
Sr.no	Symbol	Name
1		Attributes
2		Entity
3		Links (connector)
4		Relationship
5		Key attributes
6		Multi valued attributes

3.4 Entity Relationship Diagram:-

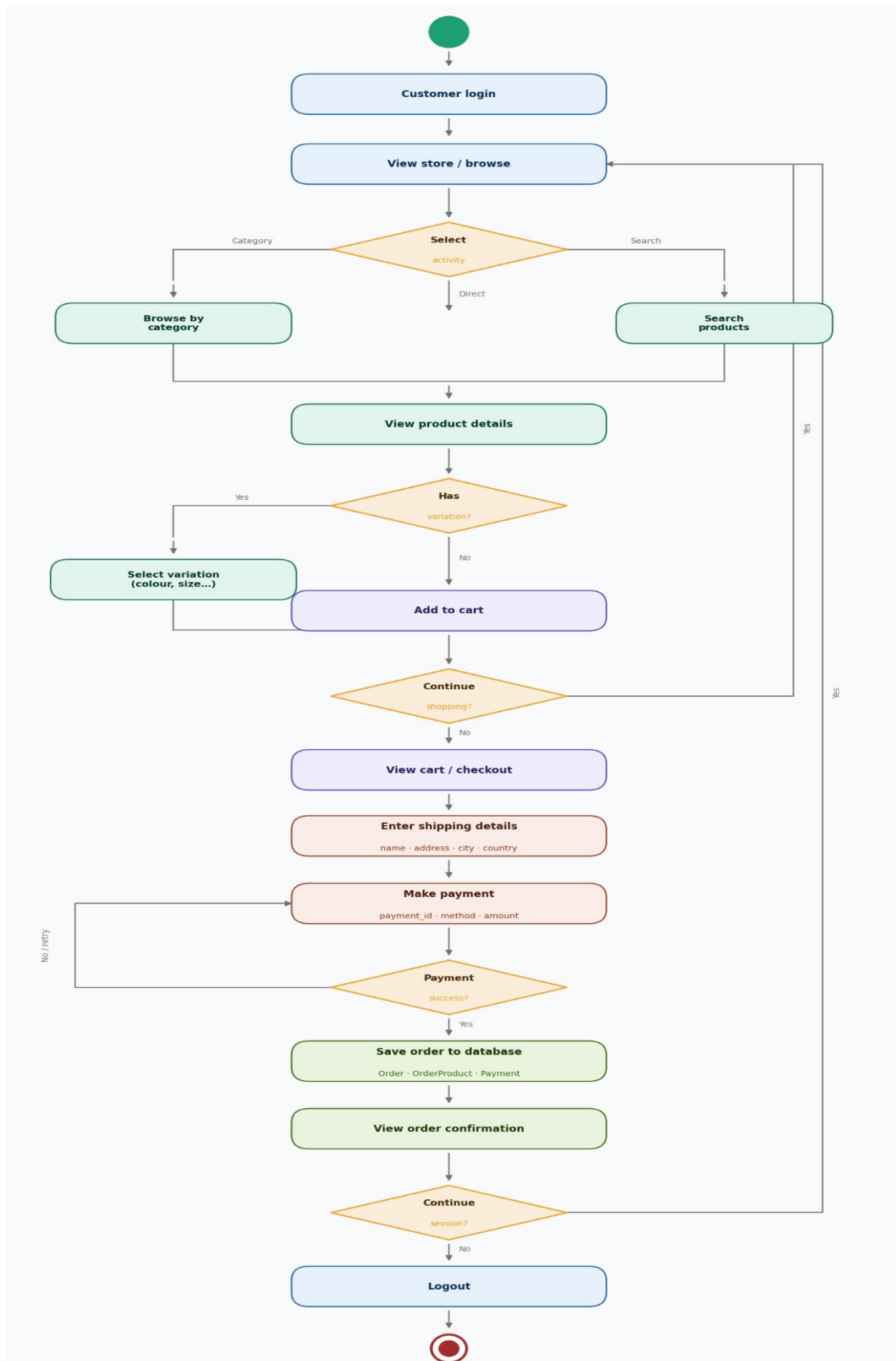


4 . UML DIAGRAM

4.1 Use Case Diagram:-



4.2 Activity Diagram:-



5. System Development

5.1 Screen Shots of Input/ Output Forms

Login Page:-

Sign in

[Forgot password?](#)

Don't have account? [Sign up](#)

Registration

Sign up

First name	Last name
Enter First Name	Enter last Name
Email Address	Phone Number
Enter Email Address	Enter Phone Number
Create password	Repeat password
Enter Password	Confirm Password

Have an account? [Log In](#)

Dashboard

GreatKart

All category

Store

Search

Q

Welcome guest!
Sign in | Register

Popular products

See all

Great-Tshirt
₹499
★★★★☆

ATX-Jeans
₹599
☆☆☆☆☆

Blue-Shirt
₹449
☆☆☆☆☆

jordan-true-flight-basketball-shoes
₹999
☆☆☆☆☆

Product details

Great-Tshirt

★★★★☆ 1 reviews

₹ 499

The Great T-shirt is stylish, comfortable, and perfect for everyday wear. It is made with soft quality fabric and gives a trendy look with a comfortable fit.

Choose Color

Select

Select Size

Select

Add to Cart

Cart

GreatKart

All category ▾Store

Search

Q

Welcome guest!
Sign in | Register

PRODUCT	QUANTITY	PRICE	
 Great-Tshirt Color : Blue Size : Small	-2+	₹ 998 ₹ 499 each	Remove
 US-Polo-Assn_Jacket	-1+	₹ 1598 ₹ 1598 each	Remove
 Blazer Color : Black Size : Large	-1+	₹ 2599 ₹ 2599 each	Remove

Total price: ₹ 5195
Tax 5% : ₹ 259.75
Grand Total: ₹ 5454.75

PayPalVISAMasterCardAmerican Express

Checkout

Continue Shopping

© 2026 HITESH

hbsatvara@gmail.com +91 90161 04143 Ahmedabad Gujarat India



Billing address

GreatKart

All category ▾Store

Search

Q

Welcome Sathvara!
Dashboard | Logout

Billing Address

First NameLast Name

EmailPhone Number

Address Line 1Address Line 2

CityStateCountry

Order Note

PRODUCT	QUANTITY	PRICE
 Great-Tshirt Color : Blue Size : Small	2	₹ 998 ₹ 499 each
 US-Polo-Assn_Jacket	1	₹ 1598 ₹ 1598 each
 Blazer Color : Black Size : Large	1	₹ 2599 ₹ 2599 each

Place Order

Continue Shopping

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Payment

Review Your Order and Make Payment

Billing Address

Sathvara Hitesh
Ahmedabad, Gujarat
Ahmedabad, Gujarat
India
hbsatvara@gmail.com
09016104143

Payment Method

Cash on Delivery / PayPal

Review Products

PRODUCT	QUANTITY	PRICE
 Great-Tshirt Color : Blue Size : Small	2	₹ 998 ₹ 499 each
 US-Polo-Assn_Jacket	1	₹ 1598 ₹ 1598 each
 Blazer Color : Black Size : Large	1	₹ 2599 ₹ 2599 each

Total price: ₹ 5195
Tax 5% : ₹ 259.75
Grand Total: ₹ 5454.75

PayPal VISA Mastercard

Cash on Delivery

Pay with PayPal

Debit or Credit Card

Powered by PayPal

Please Note: This is a demo website. Do not try to make real payments.

Payment PayPal



Pay with PayPal

Enter your email address to get started.

Email or mobile number

Forgot email?

 If you accept cookies, we'll use them to improve and customise your experience and enable our partners to show you personalised PayPal ads when you visit other sites. [Manage cookies and learn more](#)

Accept Decline

Total price:
Tax 5% :
Grand Total:

Please Note:
to make real p

1 ₹ 1598

Bill Page

GreatKart

Invoiced To
Sathvara Hitesh
Ahmedabad, Gujarat
Ahmedabad, Gujarat
India

Order #202605108
Transaction ID COD-202605108
Order Date: May 9, 2026, 8:43 p.m.
Status: COD

Products	Qty	Total
Great-Tshirt Color : Blue Size : Small	2	₹499.0
US-Polo-Assn_Jacket	1	₹1598.0
Blazer Color : Black Size : Large	1	₹2599.0
Sub Total:		₹5195.0
Tax 5% :		₹259.75
Grand Total:		₹5454.75

Thank you for shopping with us!

User Dashboard

Dashboard

My Orders

Edit Profile

Change Password

Log out

Logged in as: Sathvara Hitesh

Total Orders

5

[View Orders](#)



bca2022hitesh1705@tnraocollege.org
09016104143

Order

[Dashboard](#)[My Orders](#)[Edit Profile](#)[Change Password](#)[Log out](#)

Your order history

Order #	Billing Name	Phone	Order Total	Date
202605108	Sathvara Hitesh	09016104143	₹5454.75	May 9, 2026, 8:43 p.m.
202605107	Sathvara Hitesh	09016104143	₹5457.9	May 9, 2026, 6:59 p.m.
202605106	Sathvara Hitesh	09016104143	₹1258.95	May 9, 2026, 6:47 p.m.
202605105	Sathvara Hitesh	09016104143	₹3985.8	May 9, 2026, 6:39 p.m.
202605092	Sathvara Hitesh	09016104143	₹11773787.55	May 9, 2026, 6:05 p.m.

User profile

[Dashboard](#)[My Orders](#)[Edit Profile](#)[Change Password](#)[Log out](#)

Edit your profile



First Name

Last Name

Phone Number

Profile Picture

No file chosen

Address Line 1

Address Line 2

City

State

Country

Change password

Dashboard	Change Your Password
My Orders	Current Password
Edit Profile	
Change Password	Create new password
Log out	Confirm new password
	Submit

All catalogues

GreatKart [All category](#) [Store](#) [Welcome Sathvara!](#) [Dashboard](#) | [Logout](#) [!\[\]\(13f8cc7f38c868748825f3a80b201b57_img.jpg\)](#)

Our Store

Categories 

[All Products](#)
[T-shirt](#)
[Shirts](#)
[Jacket](#)
[Jeans](#)
[Shoes](#)
[blazer](#)

9 items found



Great-Tshirt
₹ 499

[View Details](#)



ATX-Jeans
₹ 599

[View Details](#)



Blue-Shirt
₹ 449

[View Details](#)







Search menu

GreatKart

All categoryStore

Search

Q

Welcome Sathvara!
Dashboard | Logout

Our Store

Categories

All Products
T-shirt
Shirts
Jacket
Jeans
Shoes
blazer

2 items found

Blue-Shirt
₹ 449
View Details

Wrangler-Shirt
₹ 1199
View Details

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Review page

Write Your Review

How do you rate this product?

★★★★★

Review Title:

Review:

Submit Review

2-Factor Review Submission is disabled for demo purpose.

Customer Reviews

★★★★★ 1 reviews

hitesh sathvara

★★★★★

good
best

May 9, 2026, 6:24 p.m.

6. Implementation

1 System Environment Setup

The GreatKart E-commerce project was developed in a local development environment using Django's built-in development server. A virtual environment was created to manage project dependencies and required Python packages. Different Django applications were configured for handling product management, user accounts, cart functionality, and order management. The system was tested locally to ensure proper functionality and performance.

2 Database Implementation

Django's Object Relational Mapping (ORM) was used for database design and management. Database tables were created using Django models and migrations. The database stores all information in an organised format and maintains relationships between products, categories, users, carts, and orders.

The main database tables include:

- User Authentication Table (Django default)
- Product Table
- Category Table
- Cart Table
- Cart Item Table
- Order Table

3 Implementation of E-commerce Modules

Product Module

- Displays product details with categories and pricing information
- Allows admin to add, update, and delete products

Category Module

- Organises products into different categories
- Helps users browse products easily

Cart Module

- Allows users to add products to the shopping cart
- Supports quantity management and cart updates

Order Module

- Handles order placement and order management
- Stores customer order details securely

Search Module

- Enables users to search products quickly
- Improves user experience and navigation

4 Admin Panel Implementation

- Manage user accounts
- Add and manage products
- Manage product categories
- Monitor customer orders

5 Security and Data Protection

- Password encryption
- Secure authentication system
- Session management
- User-based access control

6 User Authentication and Authorization

- Only registered users can access user features
- Users can manage their own cart and orders
- User sessions are maintained securely

7. Tools and Technologies

Tools	• Google Chrome browser • Draw.io (For Diagram) • MS PowerPoint (For presentation PPT) • MS Word (for Documentation) • Visual Studio Code
Frontend	• HTML • CSS • Bootstrap • JavaScript
Backend	• Python • Django
Database connectivity	• SQLite
Operating System	• Windows 11

8. Bibliography

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- <https://www.drawio.com>
- <https://www.youtube.com>
- <https://www.w3schools.com>
- <https://chatgpt.com/>